

USDC Savings & Global Payout Rails on Stellar

Azza's technical proposal to integrate **Blend v2** as a yield engine and **Bridge** as a Stellar-settled payout rail — adding accessible USDC savings and international (USD / EUR / GBP) settlement on top of Azza's existing custody and fiat-payout stack, delivered over WhatsApp.

DOCUMENT	DATE	TIMELINE	INTEGRATIONS
Technical Spec	June 2026	18 weeks · 3 tranches	Blend v2 · Bridge

Azza × Stellar — Integration Proposal
Consumer yield + cross-border settlement on Stellar

\$95,000
TOTAL BUILD AWARD REQUEST



Product Integration Scope

Two Stellar-native features added to an app with existing traction

Azza is a live WhatsApp-based stablecoin payments app processing thousands of monthly transactions across multiple fiat corridors. This proposal adds **two Stellar-native capabilities** on top of that traction — both reusing Azza's existing custody, KYC, and payout infrastructure rather than rebuilding it.

FEATURE 1 · YIELD

Azza Savings — WhatsApp-native USDC savings, powered by Blend v2

A savings layer that lets African consumers earn yield on idle USDC without leaving WhatsApp — no app install, no wallet setup, no DeFi knowledge required. Azza is the custodian and accounting layer; Blend v2 on Soroban is the yield engine underneath.

FEATURE 2 · SETTLEMENT

Global Payout Rails — international settlement, powered by Bridge

A new provider in Azza's payout orchestration network that uses **Bridge** — settling over Stellar — to extend on/off-ramp beyond African corridors into **USD, EUR, and GBP**. This turns Azza from an Africa-only payments app into one that can move money internationally on Stellar rails.

Who uses each

Savings serves consumers in Nigeria, Kenya, Ghana, and South Africa who already hold USDC via Azza — people receiving USDC from remittances, freelance work, or off-ramp cycles, whose balance currently sits idle. **Payout rails** serve users and businesses needing to send or receive across borders — an African freelancer paid in USD by an overseas client, or an international business funding payouts into Azza's African corridors — where Bridge provides the missing non-African leg.

The savings workflow

User journey — from a WhatsApp message to accruing yield and back to local fiat

- 1 User tells Azza via WhatsApp: Save \$50
- 2 Azza moves that USDC from the user's Azza balance to its Stellar treasury account and deposits it into a Blend v2 lending pool.
- 3 User can see their savings balance and current APY in WhatsApp at any time.
- 4 Yield accrues continuously and is reflected in the user's savings balance.
- 5 On Withdraw my savings , Azza redeems from Blend v2 — the user keeps it as USDC or off-ramps directly to their local bank or mobile money.

Infrastructure category

Two layers on the same base stack: consumer-facing **yield / savings infrastructure** (Blend v2), and **cross-border payment / settlement infrastructure** (Bridge). Neither replaces an existing rail — savings sits above the custody stack, and Bridge slots into the existing Provider Router as an additional payout provider.

5–8%

Target USDC APY via Blend v2

3+

New international corridors via Bridge (USD · EUR · GBP)

12

Fiat corridors live on Azza today

WHY THIS MATTERS FOR AZZA'S USERS

Nigeria's inflation is running above **30%**. Ghana's cedi has depreciated **40%+ in two years**. A USDC savings product earning **5–8% APY** is a concrete financial tool with no offline equivalent at this income level. At the same time, Azza's users increasingly transact with the rest of the world — and today the non-African leg of those flows is the gap. Bridge, settling on Stellar, closes it.

B Selected Stellar Integrations

Two primary building blocks from the official Integration List, plus the SDK + Horizon layer

Blend v2 PRIMARY · YIELD

What it is	Stellar's native lending protocol built on Soroban. Lenders deposit USDC into pools and earn variable APY from borrower interest.
Why this one	Blend v2 is the most mature yield source on Stellar mainnet, has SDF backing, and its USDC pools are the most liquid. The yield is real, denominated in USDC, and doesn't require Azza or its users to take on volatile-asset exposure.
Role in the stack	Azza deposits pooled user USDC into a Blend v2 USDC lending pool via Soroban contract calls. Blend v2 is the yield engine; Azza is the custodian and accounting layer above it. Users never interact with Blend directly.
Integration time	5–6 weeks — Soroban SDK, deposit / withdraw contract calls, yield-position tracking, error handling and retry logic.
External dependencies	Blend v2 Soroban contract addresses on mainnet · SDF / Blend team for contract ABI and liquidity-depth confirmation · Soroban RPC endpoint.

Bridge PRIMARY · SETTLEMENT

What it is	A stablecoin payments & treasury platform (Stripe-acquired) on the Stellar Integration List. Its Orchestration API accepts a payout instruction, then handles FX conversion, routing, and settlement — using Stellar as a settlement chain — delivering to bank rails (ACH / SEPA / SWIFT) or stablecoin wallets.
Why this one	Azza already runs a multi-provider payout network for African corridors but has no clean rail for major international currencies. Bridge fills exactly that gap: USD, EUR, and GBP settlement on Stellar through a single API, with compliance (sanctions screening, monitoring) built in — far faster than onboarding correspondent banks per currency.
Role in the stack	Bridge joins Azza's existing Offramp Provider Router as a new provider, selected for international corridors. On payout, Azza hands Bridge USDC and a destination; Bridge settles via Stellar and delivers fiat. On onramp, Bridge converts inbound USD/EUR/GBP funding into USDC credited to Azza.
Integration time	4–5 weeks — Bridge Orchestration API adapter, provider-router registration, webhook verification, Stellar settlement reconciliation, corridor/FX configuration.

External dependencies

Bridge API credentials & business onboarding · supported-corridor & settlement-asset confirmation from Bridge · Stellar settlement addresses.

Stellar SDK + Horizon **SUPPORTING**

What it is	The official Stellar JS/TS SDK and Horizon REST API for account management, transaction submission, and event monitoring. Not a named "integration partner" but a required infrastructure component.
Role in the stack	Manages Azza's Stellar treasury accounts, submits transactions to Blend v2, streams payment events via Horizon SSE, and confirms Bridge settlement transactions on Stellar.
Integration time	3–4 weeks — wallet service, key management, Horizon listener, transaction signing and submission.
External dependencies	Stellar Horizon public endpoint (SDF-hosted or self-hosted) · Soroban RPC.

Stellar-specific technical considerations

Stellar and Soroban differ from the EVM and Solana stacks Azza runs today in a few concrete ways. Each is handled within the existing service and retry architecture rather than requiring new infrastructure.

Consideration	How it's handled
Transaction & fee model	Stellar uses sequence numbers and a base reserve per account, and Soroban requires a <code>simulate-then-submit</code> step — distinct from EVM's <code>estimateGas</code> / <code>send</code> . The Wallet Service simulates each Soroban call to obtain the resource footprint and fee before signing and submitting.
Asset handling & trustlines	Holding USDC on Stellar requires a trustline to the asset. The Azza treasury account's USDC trustline is established as a one-time setup task in Tranche 1, before any deposit flow.
Treasury signer model	Single-signer treasury across Tranches 1–3 — one Azza-controlled account signs from the existing secrets manager. Multi-sig hardening is deferred post-grant; the design isolates signing so it can be upgraded without touching the flows.
Soroban-specific failure modes	Contract panics and resource exhaustion are mapped as distinct error classes inside Azza's existing retry framework, so they are caught, classified, and retried (or dead-lettered) the same way other on-chain failures already are.
Finality & monitoring	Stellar's ~5-second ledger close shapes the Horizon Listener's confirmation logic (a deposit is confirmed once its ledger closes, not optimistically). Horizon metrics and Stellar Expert provide observability into account state and transaction history.

TEAM READINESS

Tranche 1 is structured to absorb the Rust / Soroban learning curve before any mainnet exposure — all Blend v2 work is proven on testnet first, with mainnet deferred to Tranche 3. Before Tranche 2 begins, the team completes Soroban's official quickstart and a working testnet deposit/withdraw cycle as an internal readiness gate. The integration patterns themselves — contract calls, event-stream confirmation, idempotent retry handling — carry over directly from the team's existing Solana and EVM production work; what's new is the SDK and the language, not the integration discipline.

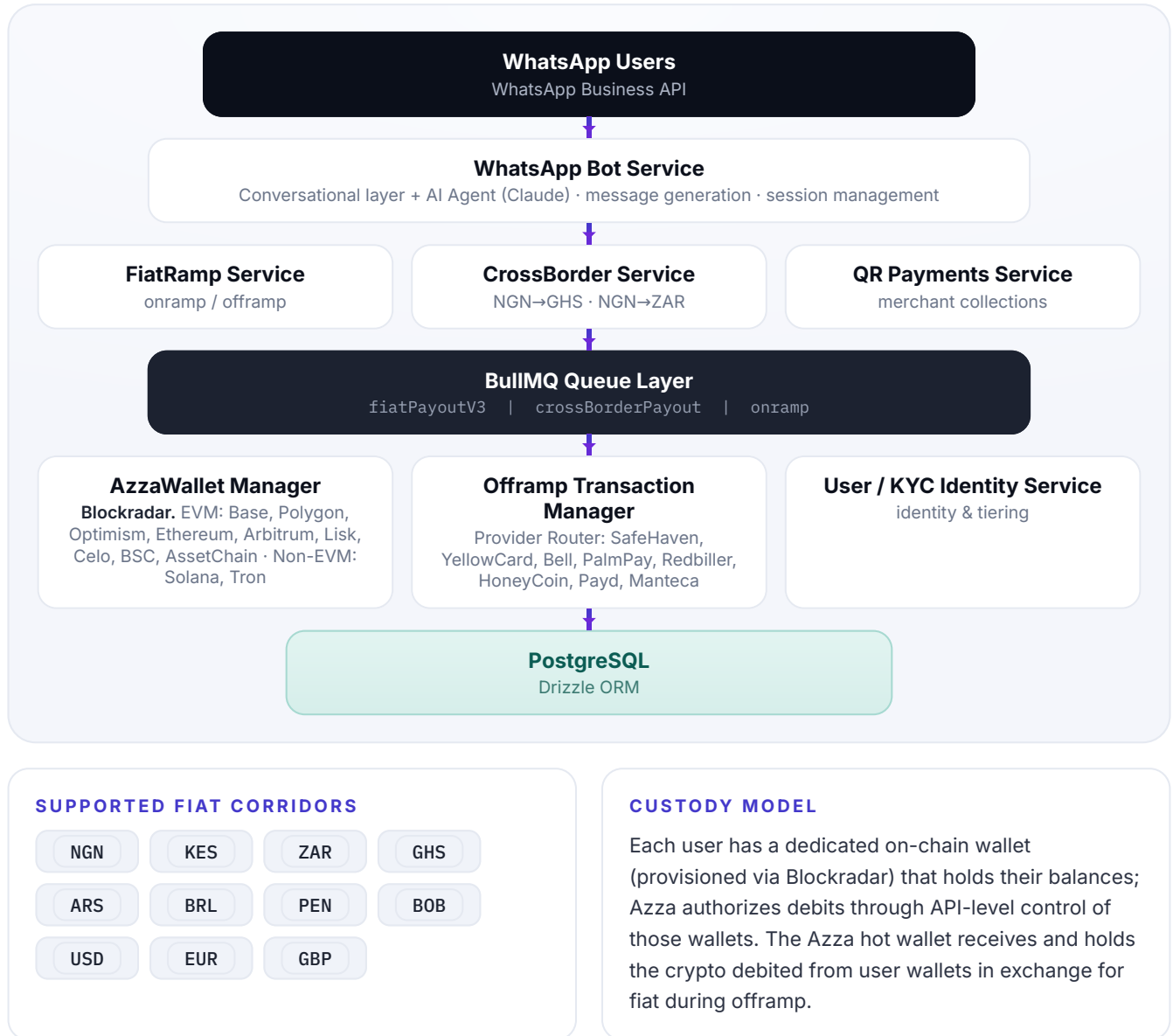


Proposed Architecture

The current Azza system, the new savings layer, and Bridge joining the payout orchestration network

TODAY · IN PRODUCTION

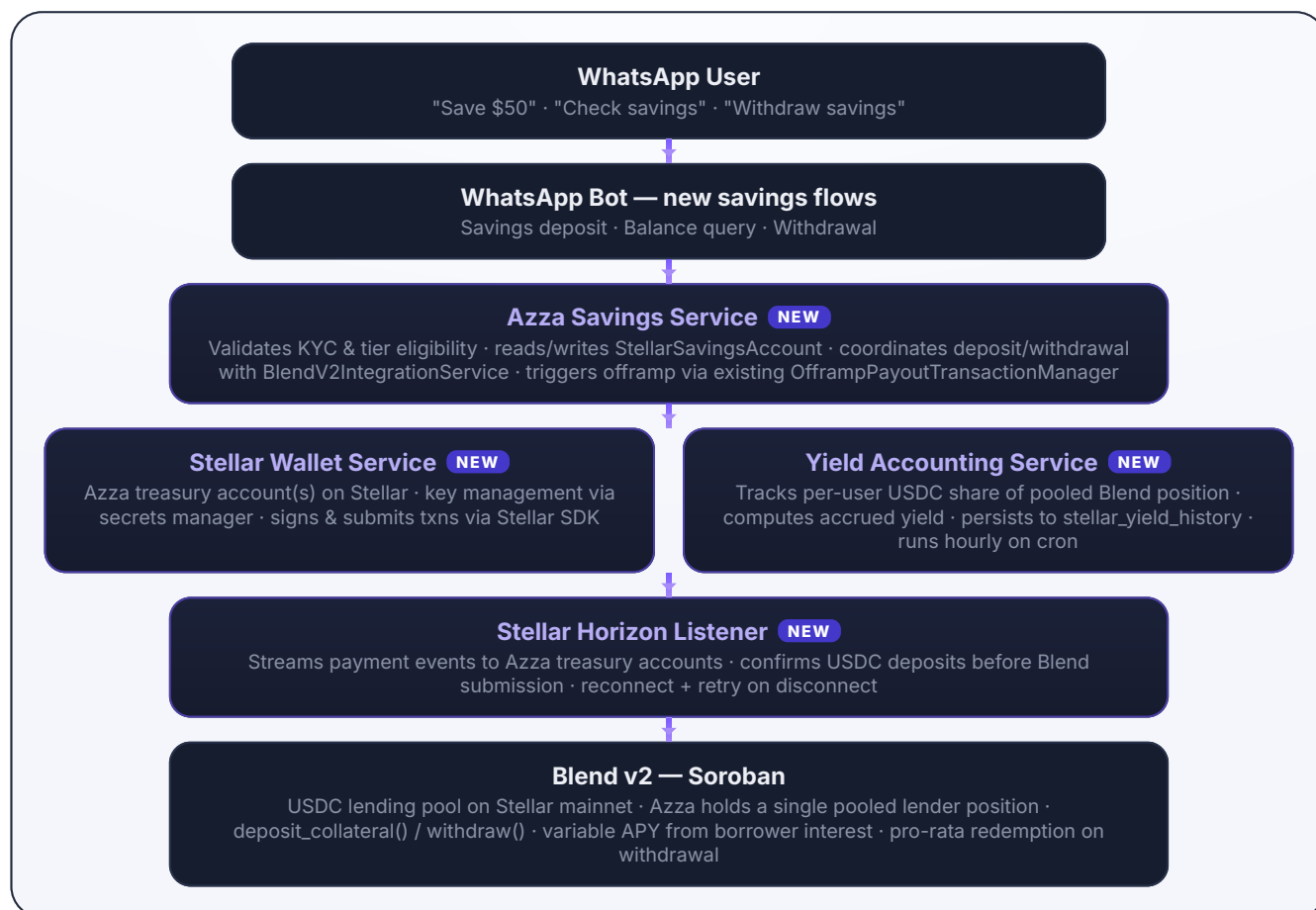
Current Azza System



NEW · THIS INTEGRATION

Azza Savings — the Stellar layer

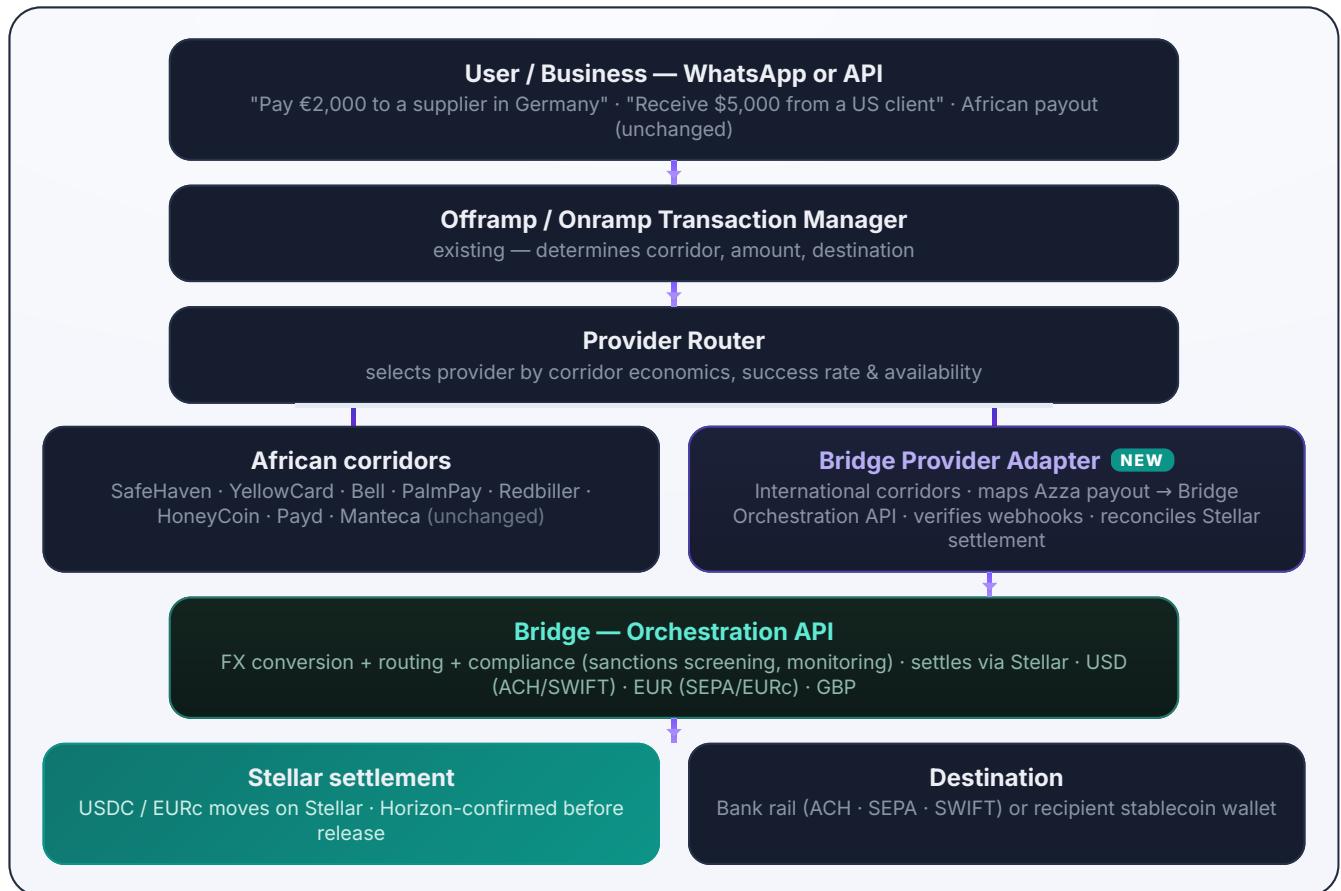
Everything below is additive. New services are marked **NEW**; on withdrawal the flow hands back into the existing offramp manager.



NEW · THIS INTEGRATION

Global Payout Rails — Bridge in the orchestration network

Azza's existing **Offramp Provider Router** already load-balances African corridors across SafeHaven, YellowCard, PalmPay and others. Bridge **NEW** joins that same router as the **international-corridor provider**, settling over Stellar — no new orchestration layer, just one more provider the router can select.



WHY THIS IS A SMALL, CLEAN INTEGRATION

Nothing in the router's existing logic changes — Bridge is registered like any other provider, and the manager keeps owning corridor selection. The new code is the Bridge adapter and Stellar settlement reconciliation. African corridors are entirely untouched, so the blast radius is contained to the new international path.

Complete product flows

Savings deposit — "Save \$50"	Savings withdrawal — "Withdraw to my bank"
> User: Save \$50 1 Savings Service validates user — KYC verified, sufficient USDC balance. 2 Debits \$50 from the user's Azza balance (internal ledger). 3 Stellar Wallet Service submits USDC transfer to the Azza treasury account. 4 Horizon Listener confirms the transfer on-chain. 5 BlendV2IntegrationService calls <code>deposit_collateral()</code>. 6 Yield Accounting records the user's pool share. ✓ WhatsApp: "Your \$50 is now earning X% APY in Azza Savings."	> User: Withdraw my savings to my bank 1 Savings Service reads savings balance + accrued yield. 2 BlendV2IntegrationService calls <code>withdraw()</code> on Blend v2. 3 Stellar Wallet Service receives USDC back to the treasury account. 4 Savings Service credits USDC to the user's Azza balance. 5 Existing OfframpPayoutTransactionManager → fiat provider → bank settlement. ✓ WhatsApp: "R850 sent to your FNB account."

International payout via Bridge — "Pay €2,000 to a supplier in Germany"
> User / business initiates a EUR payout to an international beneficiary. 1 Offramp Transaction Manager identifies the corridor as international and asks the Provider Router for a provider. 2 Provider Router selects the Bridge adapter for the EUR corridor. 3 Bridge adapter debits USDC from Azza and submits a payout instruction to the Bridge Orchestration API (destination, currency, amount). 4 Bridge converts, runs compliance screening, and settles via Stellar; the Horizon Listener confirms the settlement transaction. 5 Bridge delivers EUR to the beneficiary's bank via SEPA; the adapter reconciles the webhook against the Stellar settlement. ✓ WhatsApp: "€2,000 sent — settled on Stellar, delivered via SEPA."

Wallet handling **CUSTODIAL**

For the savings product specifically, Azza uses a **shared treasury model** — not per-user Stellar keypairs. One or a small set of Azza-controlled Stellar accounts hold the aggregate USDC deposited into Blend v2, and each user's share of that pooled position is tracked in Azza's savings ledger (`stellar_savings_accounts`). Pooling is required because Blend v2 represents Azza as a **single lender position** — there is no per-user position to hold on-chain.

Monitoring, compliance & failure handling

HOW AZZA CUSTODIES TODAY

With Blockradar, each user has their own on-chain wallet generated for them, and balances are held on-chain. Azza authorizes debits through API-level control of those wallets — it does not run an off-chain balance ledger.

WHY SAVINGS DIFFERS

A pooled Blend position can't be split into per-user on-chain wallets, so savings entitlements are tracked off-position in the savings ledger and reconciled daily against the on-chain position. Custodial: Azza holds the Stellar keys in the existing secrets manager; users interact only through WhatsApp.

Concern	Approach
Blend v2 utilisation spike pool liquidity risk	Monitor utilisation rate via a Soroban view call on a cron; auto-suspend new deposits if the rate exceeds threshold.
Horizon stream disconnect	SSE reconnect with exponential backoff; a deposit is not confirmed in Blend until the Horizon event is received.
Soroban transaction failure	Retry with an idempotency key derived from the internal deposit ID; dead-letter queue for manual review after 3 failures.
Over-withdrawal attempt	Enforced at the Savings Service layer before any Soroban call is made.
KYC gating	Savings deposit gated on <code>kycStatus === 'verified'</code> , consistent with the existing <code>YELLOWCARD_ZAR</code> rule.
Yield accounting drift	Daily reconciliation job — the sum of all user shares in the DB must equal Azza's on-chain Blend position.
Bridge settlement mismatch payout rail	Every Bridge payout reconciled against its Stellar settlement transaction and webhook; payout not marked complete until both confirm.
Bridge webhook authenticity	Signature-verify all Bridge webhooks; idempotency keys prevent duplicate payout processing on retries.
Provider failover	If Bridge is unavailable for a corridor, the Provider Router falls back per existing routing logic; international corridors with no fallback are queued, not silently failed.
Regulatory	Savings yield is earned on user-initiated deposits only, with withdrawal always available. Cross-border payouts run through Bridge's built-in compliance (sanctions screening, monitoring). Legal review recommended for Nigeria / Kenya before mainnet.

D Milestones

Three milestone tranches over 18 weeks; deliverable budgets reconcile to the disbursement split

Tranche 0 (10% / \$9,500) is released on approval as a kickoff advance. The three milestone tranches below cover the remaining 90% and map to the 20% / 30% / 40% disbursement schedule.

STELLAR ECOSYSTEM SUCCESS METRICS

By end of Tranche 3, Azza commits to routing **21–29% of total transaction volume through Stellar** (combined Bridge settlement + Blend v2 savings deposits), ramping toward a projected steady-state of **~41%** shortly after mainnet launch. This is tracked across both legs: Bridge-settled international volume (USD/GBP corridors) and cumulative USDC deposited into Blend v2 via Azza. Progress against both will be reported at the close of each tranche, not just at completion.

TRANCHE 1 Design, Infrastructure, Testnet — both integrations · Weeks 1–6

\$19,000

Deliverable	How completion is measured	Budget
Stellar treasury account + USDC transfers on testnet	Testnet USDC transfer confirmed via Horizon, balance reflected in internal ledger	\$4,500
BlendV2IntegrationService: deposit and withdraw on Blend v2 testnet	Deposit to Blend pool confirmed on-chain; withdrawal returns USDC to Azza account	\$6,000
Bridge sandbox integration: Orchestration API connected, test payout settling via Stellar	Sandbox USD/EUR payout instruction accepted; Stellar settlement tx confirmed and reconciled	\$5,000
Staging environment: Horizon listener, Savings Service, Bridge provider adapter scaffold, DB schema	Both flows runnable end-to-end in staging against testnet / sandbox	\$3,500
Tranche 1 total		\$19,000

Deliverable	How completion is measured	Budget
WhatsApp savings deposit + balance / yield display	Deposit flow tested on WhatsApp sandbox; balance and APY shown, updating within 1 hour	\$6,000
Savings withdrawal → USDC → local fiat via existing offramp	Full flow: savings withdrawal → NGN/KES bank settlement confirmed in staging	\$5,000
Bridge provider live in Provider Router: USD / EUR / GBP payout routing	Router selects Bridge for international corridors; payout settles via Stellar and delivers to bank rail in staging	\$7,000
International onramp via Bridge: USD / EUR / GBP funding → USDC	Inbound funding converted to USDC and credited to Azza balance in staging	\$3,500
Yield Accounting Service: per-user shares, APY from Blend position	Unit tests passing; simulated accrual matches expected Blend v2 rates	\$4,500
Risk monitoring: Blend utilisation guard, reconciliation, Bridge webhook verification	Alerts fire in staging on utilisation breach and on settlement / webhook mismatch	\$2,500
Tranche 2 total		\$28,500

Deliverable	How completion is measured	Budget
Blend v2 savings live on mainnet (Nigeria, Kenya)	Real user savings deposits confirmed in Blend v2 mainnet; yield accruing	\$8,000
Bridge payout rails live on mainnet: USD / EUR / GBP corridors	Real international payout settled via Stellar and delivered to a destination bank account	\$8,500
Beta cohort onboarded — operational (recruitment, support, monitoring)	Beta cohort recruited and supported; ≥80% complete at least one savings or payout cycle	\$2,000
Beta incentive pool — referral campaign for "Save on Azza" & Bridge USD payouts	Referral campaign live; sign-ups and payout usage attributed and tracked	\$2,500
Beta incentive pool — capped APY boost (first 250 beta depositors)	Boost applied to the first 250 depositors; capped spend not exceeded	\$1,500
Full QA across both integrations: edge cases, failure injection, reconciliation under load	QA report documenting all failure paths; zero silent fund loss in testing	\$7,000
Production observability: dashboards, alerts, anomaly detection	Alerts firing correctly for all monitored failure scenarios across both paths	\$5,000
Documentation, SCF reporting & handover	Architecture + runbook delivered; final SCF milestone report submitted	\$3,500
Tranche 3 total		\$38,000

Beta incentive pool, prioritized: referral campaign and capped APY boost are budgeted for this phase. Cashback and content-bounty mechanisms remain available as fallback levers if beta performance data suggests adjustment, but are not funded in this phase.



Budget

\$95,000 total · engineering-cost-based · two integrations · disbursed across four tranches

Line item	Cost	Notes
Blend v2 backend integration Soroban calls, Savings Service, Yield Accounting	\$24,000	Yield engine — savings deposit / withdraw / accrual
Bridge backend integration Orchestration API adapter, Stellar settlement reconciliation, webhooks	\$22,000	Payout rail — international corridor settlement on Stellar
Provider orchestration & payout routing Bridge registered in Provider Router, corridor / FX config	\$9,000	Wiring Bridge into the existing offramp routing layer
Wallet / asset & treasury management Stellar treasury accounts, key management, Horizon listener	\$9,000	Single-signer treasury on existing secrets manager; multi-sig hardening deferred post-grant
WhatsApp UX savings deposit, balance, withdrawal conversational flows	\$8,000	New bot flows on the existing session / message architecture
QA / testing testnet, sandbox, integration, edge cases, beta cohort	\$11,000	Both integrations — staging + mainnet beta, failure-injection coverage
Infrastructure / observability Soroban RPC, Horizon node, Bridge webhooks, staging, dashboards	\$7,000	Infra tied directly to the integrations
PM / engineering coordination milestone tracking, SCF reporting, Blend & Bridge liaison	\$5,000	~4.5-month engagement across two external dependencies
Total	\$95,000	Large scope · two integrations

Tranche disbursement

T0 · 10% on approval \$9,500	T1 · 20% testnet \$19,000	T2 · 30% core flows \$28,500	T3 · 40% mainnet launch \$38,000
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Application narrative

Azza is an active WhatsApp-based stablecoin payments app serving users across **Nigeria, Kenya, Ghana, and South Africa**, processing thousands of transactions monthly across multiple fiat corridors. We are integrating **two Stellar building blocks** on top of that traction. **Blend v2** adds a savings layer — users deposit USDC into yield-bearing positions from WhatsApp, earn variable APY, and withdraw to local fiat. **Bridge** joins our existing payout orchestration network as a Stellar-settled provider for international corridors (USD, EUR, GBP), extending Azza beyond African rails. Both are technically grounded: Azza already custodies USDC, runs KYC, and operates a multi-provider payout router — Blend adds the yield engine and Bridge adds the international settlement leg. For Stellar, this brings a non-crypto-native African user base to Soroban and routes real cross-border volume across Stellar settlement, through a distribution channel with proven traction.

At full run-rate, Azza projects **~\$300,000/month** in Bridge-settled USD/GBP volume plus **~\$80,000/month** from Blend v2 savings adoption — together **~41% of total transaction volume** routed through Stellar at steady state.

Savings turns idle stablecoins into income; Bridge turns Azza into a rail that reaches the rest of the world — both running on Stellar.